

The Energy Balance Project

Discrete Mathematical Foundation — Version 2.8

A synchronous discrete energy–dissipation inequality for Model D0, with explicit finite-step remainder

Status: mathematics only, NOT YET PEER REVIEWED. Every derivation in this note is a proof attempt awaiting independent expert review. No physical engine, test of prior versions, or EBU-accounting behaviour is changed by this stage.

Numerical validation is not mathematical proof. The companion `test_v28.py` (11 groups, 132 numerical checks, standard library only, deterministic seed 20260726) validates the statements below on declared fixtures, deterministic samples, and negative controls. A passing run means the claims hold at the tested points; it establishes no theorem, and the check count is not a count of theorems.

Scope in one sentence. V2.7 proved a *continuous-time* energy–dissipation identity for the smooth, unconstrained, simultaneous Onsager flow (Model C). V2.8 derives a *discrete-time* counterpart for the simplest compatible synchronous system (**Model D0**: frozen-state, simultaneous, unconstrained, loss-aware explicit Euler) — and states precisely why that result does **not** yet cover the production Python engine family (the DE family, §3). Proving D0 is a first brick, not the building.

Epistemic labels are used strictly: **Definition / Assumption / Lemma / Theorem / Corollary / Proof / Counterexample / Conjecture / Numerical validation**. This PDF is generated from `Foundation_v2.8_discrete_draft.md` (the authoritative source) by `make_paper_v28_discrete.py`; equations are typeset with the matplotlib `mathtext` engine.

1. Purpose and relationship to V2.7

V2.7 (Theorem 7.1) established, in continuous time, for the derived Onsager flow $dx/dt = u(x) + S J(x)$:

$$\frac{dV}{dt} = \sum_i \mu_i u_i - \sum_e \left(\frac{J_e^2}{M_e} + \theta_e J_e \right) \leq \sum_i \mu_i u_i$$

The Python engine does not run this flow. It advances in ticks of size $\Delta t = 1$, applies **natural drive and transport as two ordered sub-steps**, evaluates whatever transport force it computes **loss-blind** ($\mu_i - \mu_j - \theta$; the forced harness of §3 computes none), applies transfers **sequentially against live state**, and **clips** to $[0, K]$. Theorem 7.1 explicitly excluded all of that.

This note takes the first step across that gap: a **synchronous, unconstrained, loss-aware, explicit-Euler** discretisation (Model D0), for which we prove a discrete descent inequality with an *explicit* finite-step remainder, a one-edge sufficient step-size condition, conservative graph-level conditions (spectral, active-set, and state-specific), a conservation/loss ledger, and a one-tick locality theorem — while cataloguing, with counterexamples, exactly where the result stops.

2. Notation and standing assumptions

Definition 2.1 (state and functional). State $x \in \mathbb{R}^n$ (unconstrained here — no $[0, K]$ box; see §11). Separable state functional and local potential:

$$V(x) = \sum_i v_i(x_i), \quad \mu_i(x) = \frac{\partial V}{\partial x_i} = v'_i(x_i), \quad \mu(x) = \nabla V(x)$$

For the burden functional of the project (viable band $[L_i, U_i]$, regenerative reserve R_i):

$$v_i(x) = \alpha_i [L_i - x]_+^2 + \beta_i [x - U_i]_+^2 + \chi_i [R_i - x]_+^2$$

so ∇V is piecewise-linear and continuous and $V \in C^1$ (each $v_i \in C^1$, v_i '' a step function).

Definition 2.2 (directed lossy edge). For an edge $e = (i, j)$ with efficiency $\eta_e \in [0, 1]$, the state-change column $S_e \in \mathbb{R}^n$ has -1 in slot i and $+\eta_e$ in slot j ; a scalar transfer q_e maps $x_i \rightarrow x_i - q_e$, $x_j \rightarrow x_j + \eta_e q_e$. Stacking columns into S and fluxes into J , total transport is $S J = \sum_e J_e S_e$.

Definition 2.3 (loss-aware force and Onsager flux). The force is the negative directional derivative of V along the transfer direction; with threshold $\theta_e \geq 0$ and mobility $M_e > 0$ the raw Onsager flux follows:

$$f_e(x) = -\nabla V(x)^\top S_e = \mu_i - \eta_e \mu_j, \quad J_e(x) = M_e [f_e(x) - \theta_e]_+$$

The sign/convention is *derived*, not assumed; Counterexample D (§10) shows $\mu_i - \mu_j$ is the **wrong** force when $\eta_e < 1$.

Definition 2.4 (natural drive). $u_i(x) = s_i + g_i(x_i) - d_i - \lambda_i - \kappa_i x_i$, collecting supply, regeneration, demand, constant and proportional leak (unconstrained: no saturation or clipping here).

Assumption 2.5 (L-smoothness of V, with the CORRECTED constant). ∇V is L_V -Lipschitz in the Euclidean norm. The exact branchwise slope of the gradient is

$$v'_i(x) = -2\alpha_i(L_i - x)_+ + 2\beta_i(x - U_i)_+ - 2\chi_i(R_i - x)_+$$

$$v_i''(x) = 2 [\alpha_i 1_{x < L_i} + \beta_i 1_{x > U_i} + \chi_i 1_{x < R_i}]$$

Because the homeostatic deficit branch ($x < L_i$) and the reserve branch ($x < R_i$) — or the excess branch ($x > U_i$) and the reserve branch — can be **active simultaneously**, the slope is a **sum** of active weights, not their maximum. The exact global Lipschitz constant and its convenient safe upper bound (using $L_i \leq U_i$, so at most one homeostatic branch is active) are:

$$L_V = \max_i \sup_x 2 [\alpha_i 1_{x < L_i} + \beta_i 1_{x > U_i} + \chi_i 1_{x < R_i}] \leq 2 \max_i [\max(\alpha_i, \beta_i) + \chi_i]$$

The former draft's constant $2 \max_i \max(\alpha_i, \beta_i, \chi_i)$ is **too small** whenever a homeostatic and the reserve penalty overlap — see Counterexample E (§10). The symbol L_V is used consistently in every descent, one-edge, graph, and driven bound below. All norms are Euclidean; the matrix norm is the spectral norm.

3. Three models kept strictly separate

Definition 3.1 (Model C — continuous Onsager law, V2.7). $dx/dt = u(x) + S J(x)$, all quantities at the instantaneous $x(t)$; smooth, unconstrained, simultaneous. Subject of V2.7 Theorem 7.1.

Definition 3.2 (Model D0 — ideal synchronous discrete law, THIS note). Frozen-state, simultaneous, explicit (forward) Euler discretisation of C:

$$x^{n+1} = x^n + \Delta t (u(x^n) + S J(x^n)) \quad (\text{D0})$$

All forces and fluxes are evaluated from the single state x^n ; no clipping, no constraints, loss-aware force (Def 2.3). This is the subject of the new theorems.

Definition 3.3 (the DE engine family — three distinct update laws). The Python operators actually run are **not one identical map**; they share related state accounting but differ. What all three share is only this: a natural-update stage N applied **first**, followed by **sequentially applied actions** against live state, at fixed $\Delta t = 1$. Whether — and how — a force is computed differs per member: **DE-core** computes proposals from μ at the post-drive state $y = N(x^n)$ using the loss-blind force $F = \mu_i - \mu_j - \theta$; the **DE24-family** (six rules) computes proposals from marginals m at $y = N(x^n)$ with the same loss-blind destination term ($F = m_i - m_j - \theta$, no η weight on m_j ; reserve-aware modes add the on-site reserve marginal to m); **DE26-forced** computes **no force and no proposal** — it executes externally supplied actions subject only to feasibility caps.

Member	Function	Transport sizing	Conflict scaling	Constraints
DE-core	energy_balance.step	raw $q = M[F]_+$	proportional source scaling	feasibility caps + clip to $[0, K]$
DE24-family	ebu_v24.step_v24 (6 rules)	rule-dependent: golden-section line search or horizon-optimised q ; gated by strict decrease or horizon impact	none (sequential, sorted by F)	per-proposal q_{hi} ; hard reserve floor in hard_reserve only
DE26-forced	ebu_v26.forced_tick	externally supplied quantities; no force or search computed	none	feasible_q caps (q_{max} , source floor, dest headroom)

Assumption 3.4 (non-transfer of results). A theorem about **D0** does **not** automatically hold for any member of the DE family. **D0 is the forward-Euler discretisation of Model C; the DE members are not.** The loss-blind raw law (DE-core) and the approximate coordinate-search rules (DE24-family) are not faithful discretisations of the loss-aware Onsager law C — their proposal forces are loss-blind (Counterexample D) and, for the DE24-family, the sizing rule differs entirely; DE26-forced computes no force at all — it is an execution harness for arbitrary feasible actions. D0 and the DE family differ in at

least: operator splitting; the state at which the proposal force is frozen (x^n vs $N(x^n)$); loss-aware vs loss-blind proposal force vs no force; simultaneous vs sequential application; conflict scaling (DE-core) or line-search/horizon sizing (DE24-family); unconstrained vs clipped; and Δt free vs $\Delta t = 1$.

4. One-edge derivation: exact identities and the descent inequality

4.1 Exact first-order identity

Lemma 4.1 (first-order contribution). For the D0 update, writing $\Delta x = x^{n+1} - x^n = \Delta t(u + SJ)$ (all at x^n),

$$\nabla V(x^n)^\top \Delta x = \Delta t \nabla V(x^n)^\top u(x^n) - \Delta t \sum_e f_e J_e$$

Proof. $\nabla V^\top \Delta x = \Delta t(\nabla V^\top u + \nabla V^\top SJ)$. Per edge, $\nabla V^\top S_e = -f_e$ by Def 2.3, so $\nabla V^\top SJ = -\sum_e f_e J_e$. ■

Lemma 4.2 (edge dissipation identity). For every edge,

$$f_e J_e = \frac{J_e^2}{M_e} + \theta_e J_e$$

Proof. If the edge is **active** ($f_e > \theta_e$, so $J_e = M_e(f_e - \theta_e) > 0$), then $f_e = J_e/M_e + \theta_e$; multiply by J_e . If **inactive** ($f_e \leq \theta_e$, so $J_e = 0$), both sides are 0 regardless of the sign of f_e . ■

Combining, the first-order transport term is $-\Delta t \sum_e (J_e^2/M_e + \theta_e J_e) \leq 0$ (each summand ≥ 0 since $J_e \geq 0$, $\theta_e \geq 0$).

4.2 Discrete descent inequality with explicit remainder

Lemma 4.3 (descent lemma). Under Assumption 2.5, for any x, y in the region: $V(y) \leq V(x) + \nabla V(x)^\top (y - x) + (L_V/2) \|y - x\|^2$.

Theorem 4.4 (one-step discrete inequality). Under Assumption 2.5, the D0 update satisfies, with $\Delta x = \Delta t(u + SJ)$:

$$V(x^{n+1}) - V(x^n) \leq \Delta t \nabla V(x^n)^\top u(x^n) - \Delta t \sum_e \left(\frac{J_e^2}{M_e} + \theta_e J_e \right) + R_n \quad (\star)$$

$$R_n := \frac{L_V}{2} \|\Delta x\|^2 = \frac{L_V \Delta t^2}{2} \|u + SJ\|^2 = \frac{L_V \Delta t^2}{2} (\|u\|^2 + 2u^\top SJ + \|SJ\|^2)$$

The remainder splits explicitly, exposing the **drive-transport cross term** $2u^\top SJ$. **Proof.** Apply Lemma 4.3 with $x = x^n$, $y = x^{n+1}$; substitute Lemma 4.1 for the first-order term and Lemma 4.2 for the edge sum; expand $\|u + SJ\|^2$. ■

Remark 4.5 (remainder notation — two objects, kept apart). The **exact Taylor remainder** of the functional along the step is

$$r_n := V(x^{n+1}) - V(x^n) - \nabla V(x^n)^\top \Delta x, \quad |r_n| \leq \frac{L_V}{2} \|\Delta x\|^2 = R_n$$

(the two-sided quadratic bound for an L_V -Lipschitz gradient). R_n **equals** its expression by definition — it is not itself 'bounded by' it — and $R_n = O(\Delta t^2)$. Neither r_n nor R_n is the local truncation error of the state trajectory; §7 keeps the three error notions separate.

5. Step-size bounds: one edge, graph, active set, state-specific

5.1 Undriven descent, one edge

Theorem 5.1 (one-edge step-size bound, $u = 0$). For a single active edge $e = (i, j)$ with $u \equiv 0$, the D0 step satisfies $V(x^{n+1}) \leq V(x^n)$ whenever

$$\Delta t \leq \frac{2}{L_V M_e (1 + \eta_e^2)}$$

Including θ_e only relaxes this: the true admissible range is $\Delta t \leq (2/(L_V(1+\eta_e^2)))(1/M_e + \theta_e/J_e)$. **Proof.** With $u = 0$, (★) gives $V(x^{n+1}) - V(x^n) \leq -\Delta t (J^2/M_e + \theta_e J) + (L_V \Delta t^2/2) \|S_e\|^2 J^2$ with $\|S_e\|^2 = 1 + \eta_e^2$; drop the non-negative $\theta_e J$ term and solve. ■

Consistency with V2.7 §5. Setting $\Delta t = 1$ gives $M_e \leq 2/(L_V(1+\eta_e^2))$; with $L_V = 2w$ (a single homeostatic branch of weight w , no reserve overlap) this is $M_e \leq 1/(w(1+\eta_e^2))$ — identical to the V2.7 §5.2 symmetric single-transfer bound. Counterexample A (§10) shows the bound is **tight** (necessary and sufficient) in the symmetric *pure-quadratic* fixture defined there.

5.2 Undriven descent, graph

Theorem 5.2 (spectral step-size bound, $u = 0$). Let $D_M = \text{diag}(M_e)$. If

$$\Delta t \leq \frac{2}{L_V \|S D_M^{1/2}\|_2^2}$$

then the D0 step satisfies $V(x^{n+1}) \leq V(x^n)$ for $u = 0$. The matrix inequality in the proof is **uniform over all vectors J** , so in particular it holds for every **Onsager-generated** flux vector (Def 2.3); the descent conclusion still relies on that flux law (via Lemma 4.2), not on an arbitrary J .

Proof. From (★) with $u = 0$, after dropping the non-negative θ term, a sufficient condition is $J^T D_M^{-1} J \geq (L_V \Delta t/2) J^T S^T S J$. Substituting $y = D_M^{-1/2} J$ this reads $\|y\|^2 \geq (L_V \Delta t/2) y^T (D_M^{-1/2} S^T S D_M^{-1/2}) y$ for all y , i.e. $(L_V \Delta t/2) \lambda_{\max}(D_M^{-1/2} S^T S D_M^{-1/2}) \leq 1$. Since $D_M^{-1/2} S^T S D_M^{-1/2} = (S D_M^{1/2})^T (S D_M^{1/2})$, its largest eigenvalue is the squared spectral norm; rearrange. Uniformity over J is a convenience — it does **not** assert that an arbitrary flux vector dissipates; only the Onsager-law flux does, through Lemma 4.2. ■

Corollary 5.3 (one-edge special case). For a single edge $\|S D_M^{1/2}\|^2 = M_e(1 + \eta_e^2)$, recovering Theorem 5.1 exactly.

Remark 5.4 (structure of $S^T S$, and a degree-weighted Gershgorin bound). $(S^T S)_{ee'} = S_e \cdot S_{e'}$: diagonal $1 + \eta_e^2$; +1 for two edges sharing their **source**; $\eta_e \eta_{e'}$ for two sharing their **destination**; $-\eta_e$ when one edge's destination is the other's source. Using the **similarity** (equal spectra) $D_M^{-1/2} S^T S D_M^{-1/2} \sim D_M^{-1} S^T S$ and applying Gershgorin to the rows of the (generally non-symmetric) $D_M^{-1} S^T S$:

$$\|S D_M^{1/2}\|_2^2 = \lambda_{\max}(D_M S^T S) \leq \max_e M_e \left[(1 + \eta_e^2) + \sum_{e' \neq e} |S_e \cdot S_{e'}| \right]$$

Theorem 5.5 (active-set spectral bound, $u = 0$). Let $A(x^n) = \{ e : f_e(x^n) > \theta_e \}$ be the active edge set, and $S_A, D_{M,A}$ the restrictions to it. Because inactive edges carry $J_e = 0$, the proof of Theorem 5.2 applies verbatim restricted to A . Hence

$$\Delta t \leq \frac{2}{L_V \|S_A D_{M,A}^{1/2}\|_2^2}$$

guarantees $V(x^{n+1}) \leq V(x^n)$ for the current step. Since $A(x^n) \subseteq E$, the active-set norm never exceeds the full norm, so Theorem 5.5 is never worse than Theorem 5.2 and often strictly better.

Lemma 5.6a (the Onsager flux has no nonzero null flow). At any state, the Onsager flux J of Def 2.3 satisfies

$$\mu^\top SJ = - \sum_e \left(\frac{J_e^2}{M_e} + \theta_e J_e \right)$$

Consequently $SJ = 0 \Rightarrow J = 0$. **Proof.** By the per-edge computation of Lemma 4.1, $\mu^\top S_{\cdot e} = -f_{\cdot e}$, so $\mu^\top SJ = -\sum_e f_{\cdot e} J_e$; by Lemma 4.2 (valid precisely because J is the Onsager flux), $f_{\cdot e} J_e = J_e^2/M_e + \theta_e J_e$ on every edge. If $SJ = 0$ the left side is 0, so the nonnegative sum vanishes term by term; since $M_e > 0$, $J_e^2/M_e = 0$ forces $J_e = 0$ for all e . ■

Scope. An **arbitrary externally prescribed** flow may of course satisfy $SJ = 0$ with $J \neq 0$ — e.g. an equal circulation around a lossless cycle — but such a flow is **not generated by the Onsager law** of Def 2.3 and lies outside Lemma 5.6a and Theorem 5.6 (cf. the uniformity remark in Theorem 5.2: only the Onsager-law flux dissipates through Lemma 4.2).

Theorem 5.6 (direct state-specific bound, $u = 0$). At a fixed state x^n with Onsager flux $J = J(x^n)$: if $J = 0$ (no active edge), the transport step is trivial and V is unchanged by transport; if $J \neq 0$, then necessarily $SJ \neq 0$ (Lemma 5.6a), and

$$\Delta t \leq \frac{2 \sum_e (J_e^2/M_e + \theta_e J_e)}{L_V \|SJ\|^2}$$

guarantees $V(x^{n+1}) \leq V(x^n)$ (the bound is well defined and strictly positive: $\|SJ\| > 0$ and the dissipation sum is > 0 whenever $J \neq 0$). **Proof.** Direct from (★) with $u = 0$: if $J = 0$ the whole transport term vanishes; if $J \neq 0$, Lemma 5.6a gives $\|SJ\| > 0$, and solving the quadratic-in- Δt inequality for a nonpositive right-hand side yields the stated bound. ■

Conjecture 5.7 (tightness — OPEN). The active-set (5.5) and state-specific (5.6) bounds are *sufficient* current-step conditions. Whether either coincides with the **tight** admissible Δt is open; the pure-quadratic one-edge fixture of Counterexample A is the only case where tightness is established. No empirical threshold is claimed as a theorem.

6. The discrete driven inequality ($u \neq 0$)

Theorem 6.1 (driven one-step inequality). Under Assumption 2.5, (★) holds verbatim for $u \neq 0$. V is governed by the competition of three terms — drive, dissipation, remainder:

$$V(x^{n+1}) - V(x^n) \leq \Delta t \mu^\top u - \Delta t \sum_e \left(\frac{J_e^2}{M_e} + \theta_e J_e \right) + \frac{L_V \Delta t^2}{2} \|u + SJ\|^2$$

(first term: **drive**; second: **dissipation**; third: **finite-step remainder**.)

Corollary 6.2 (sufficient one-step decrease). $V(x^{n+1}) \leq V(x^n)$ holds if

$$\sum_e \left(\frac{J_e^2}{M_e} + \theta_e J_e \right) \geq \mu^\top u + \frac{L_V \Delta t}{2} \|u + SJ\|^2$$

Two readings: (i) if $\mu^\top u \leq 0$ (drive already lowers V) and Δt is small enough for dissipation to dominate the remainder, V decreases; (ii) if $\mu^\top u > 0$ (drive raises V), decrease requires transport dissipation to exceed the drive **plus** the remainder.

Non-result 6.3. V is **not** monotone in the driven case in general — Counterexample C (§10) exhibits a drive making V strictly increase in one step. We claim only the conditional Corollary 6.2, never unconditional descent.

7. Relationship to V2.7 (Model C)

Consistency ($\Delta t \rightarrow 0$) — from the exact identity, not the inequality. Because V is C^1 , the directional derivative is exact; applying the **exact** Lemma 4.1 and Lemma 4.2 gives $dV/dt = \mu^T u - \sum_e (J_e^2/M_e + \theta_e J_e)$, which is exactly V2.7's Theorem 7.1. Equivalently the D0 forward difference converges to it because $R_n/\Delta t = (L_V \Delta t/2) \|u + SJ\|^2 \rightarrow 0$. Theorem 7.1 is recovered from the exact identity plus differentiability, with the inequality only bounding the finite- Δt gap.

Three distinct error quantities (do not conflate). (1) the **exact Taylor remainder r_n of V** (Remark 4.5), rigorously bounded by $|r_n| \leq R_n = (L_V \Delta t^2/2) \|u + SJ\|^2 = O(\Delta t^2)$; (2) the **local truncation error of the state trajectory**, $x(t_n + \Delta t) - x^{n+1} = O(\Delta t^2)$ — valid only in smooth regions away from the switching surfaces $\{x_i = L_i, u_i, R_i\}$ where ∇V is only Lipschitz, not C^1 ; (3) the **accumulated global trajectory error** $O(\Delta t)$ over a fixed interval, which additionally needs a locally Lipschitz field and a bounded solution. Item (1) is about the functional V ; items (2)–(3) are about the trajectory; they are different objects.

Finite- Δt trajectories are not identical to C. The Euler iterate departs from the continuous flow by the accumulated $O(\Delta t)$ error; equality holds only in the limit.

The safe search is an approximate minimiser, not the Onsager flux. In the engine family, the line-sized rules of the DE24-family size a transfer by `_golden_min`, a finite 24-iteration golden-section search — an *approximate bounded one-dimensional minimiser* of $q \rightarrow v_i(x_i - q) + v_j(x_j + \eta q)$ on $[0, q_{hi}]$, gated on strict decrease (the horizon-sized rules optimise a different, horizon-impact objective again). This is an (approximate) coordinate-descent-style step on V , **not** the explicit flux $q = M_e[f_e - \theta_e]_+$, and it is **not** proximal (no proximal objective or proof is claimed).

Proving D0 does not prove the DE family. Beyond the flux-vs-search point, the DE members differ from D0 by operator splitting, the proposal state frozen at $N(x^n)$ (where a proposal is computed at all), the loss-blind proposal force (DE-core, DE24-family; DE26-forced computes no force), sequential live state, conflict scaling (DE-core), and clipping (§3, §10 B and D). Each is out of scope (§11).

8. Conservation and loss ledger

Theorem 8.1 (synchronous stock balance). With $\mathbf{1}$ the all-ones vector, the D0 step changes total stock by

$$\mathbf{1}^T (x^{n+1} - x^n) = \Delta t \left[\mathbf{1}^T u(x^n) - \sum_e (1 - \eta_e) J_e(x^n) \right]$$

Proof. $\mathbf{1}^T \Delta x = \Delta t (\mathbf{1}^T u + \mathbf{1}^T S J)$ and $\mathbf{1}^T S_e = (-1) + \eta_e = -(1 - \eta_e)$. ■

Corollary 8.2 (loss is explained, not destroyed). Since $\eta_e \leq 1$, $J_e \geq 0$, transport removes exactly $\sum_e (1 - \eta_e) J_e \geq 0$ units of stock — the **efficiency loss**, an accounted outflow. Lossless edges ($\eta_e = 1$) conserve stock exactly; all stock change is then attributable to the natural drive.

Remark 8.3 (two different statements). Stock balance (Theorem 8.1, about $\mathbf{1}^T x$) and Lyapunov descent (Theorems 4.4/6.1, about V) are independent: V is not a stock, and a lossy transfer that *reduces* V still *removes* physical stock. Keep them separate.

9. Locality and causal speed

Definition 9.0 (undirected dependency graph). G_u joins i and j iff a directed transfer edge (i,j) or (j,i) exists; dist is graph distance in G_u . **Assumption 9.0' (on-site drive).** Each $u_i(x)$ depends only on x_i (Def 2.4). A drive coupling several cells would break the result below.

Theorem 9.1 (one-tick dependency radius, D0). Under the synchronous frozen-state update with Assumption 9.0', x^{n+1}_i depends only on $\{x^n_k : \text{dist}(i,k) \leq 1\}$. By induction, x^{n+m}_i depends only on cells within distance m : information propagates at most one edge per tick. **Proof.** u_i is on-site (0 hops); each incident $J_e = M_e[\mu_a - \eta_e \mu_b - \theta_e]_+$ depends only on the two

endpoints of e (1 hop). Compose and induct. ■

Counterexample / Observation 9.2 (sequential live state breaks this). Every member of the DE family applies its transfers (accepted proposals, or externally supplied actions in DE26-forced) **sequentially against live state**. Then a transfer on (i,j) mutates x_j *before* a later transfer on (j,k) reads it, so x^{n+1}_k can depend on x^n_i — a **2-hop** influence in a single nominal tick (V2.7 §6 measured +0.124 at distance 2 on a $0 \rightarrow 1 \rightarrow 2$ chain, vs exactly 0 under frozen-state simultaneous application). Theorem 9.1 holds for **D0 only**; the DE family's causal speed per tick is bounded by the longest chain of transfers sharing cells in application order, not by 1.

10. Counterexamples (necessity witnesses)

Counterexample A (step above the bound increases V; tightness). Pure-quadratic fixture with no flat band: $L_i = U_i = L_j = U_j = 0$, $\chi = 0$, equal weights $\alpha = \beta = w$, so $v(x) = w x^2$ globally and $L_V = 2w$. One edge, $\eta = 1$, $\theta = 0$, state $(d, -d)$. Then $f = 4wd$, $J = 4Mwd$, and exactly

$$V_{\text{after}} = 2wd^2 (1 - 4Mw \Delta t)^2 \Rightarrow V_{\text{after}} > V_{\text{before}} \iff \Delta t > \frac{1}{2Mw}$$

The Theorem 5.1 bound here is $2/(2w \cdot M \cdot 2) = 1/(2Mw)$ — **exactly tight** on this fixture: any Δt above it strictly increases V (e.g. $w = M = 1$, $d = 1$, $\Delta t = 0.6$: V goes $2 \rightarrow 3.92$). Tightness is claimed for this fixture only; a positive-width viable band makes the bound merely sufficient (Conjecture 5.7).

Counterexample B (sequential 3-cell over-propagation). See Observation 9.2: the $0 \rightarrow 1 \rightarrow 2$ chain under sequential live-state application transmits a cell-0 perturbation to cell 2 within one nominal tick, exceeding the one-edge-per-tick propagation Theorem 9.1 proves for the synchronous law. This invalidates any attempt to extend Theorem 9.1 to the DE family unchanged.

Counterexample C (drive increases V). A single supplied cell c with $v_c(x) = \beta[x - U]^2_+$, no active transport edges, initial state $x^n_c = U$ (so $\mu_c = 0$), constant supply $s > 0$. The D0 step gives $V(x^{n+1}) - V(x^n) = \beta(\Delta t - s)^2 > 0$. Driven V is not monotone in general (supports Non-result 6.3); only the conditional Corollary 6.2 survives.

Counterexample D (the loss-blind force is wrong for $\eta < 1$). Take $\eta_e = 0.5$ and endpoint potentials $\mu_i = -3$, $\mu_j = -4$ (both cells in deficit, j more deficient). The **loss-blind** rule sees $\mu_i - \mu_j = 1 > 0$ and would transfer $i \rightarrow j$. But the **true** loss-aware force is $f_e = -3 - 0.5(-4) = -1 < 0$: by Lemma 4.1 the first-order change from this edge is $-f_e(\Delta t J) = +\Delta t J > 0$ — the loss-blind transfer **increases** V , because at $\eta = 0.5$ the efficiency loss wastes more than the deficit relief it buys. This criticism applies exactly to the members that **compute** a loss-blind proposal force — DE-core and the DE24-family — and is why D0 must use the loss-aware force. DE26-forced computes no force, so this particular criticism does not apply to it.

Counterexample E (the former Lipschitz constant is too small). A counterexample to the *former* constant $2 \max_i \max(\alpha, \beta, \chi)$, not to the corrected theorem. Take $L = R = 0$, $\alpha = \chi = 1$ (β inactive), so for $x < 0$: $v(x) = 2x^2$, $v' = 4$. One edge, $\eta = M = 1$, $\theta = 0$, $\Delta t = 0.4$, states $(-1, -2)$. Then $\mu = (-4, -8)$, $f = 4$, $J = 4$, the step maps $(-1, -2) \rightarrow (-2.6, -0.4)$, and V **increases**: $10 \rightarrow 13.84$. The former constant would read $L = 2$, whose one-edge bound $\Delta t \leq 0.5$ *incorrectly permits* $\Delta t = 0.4$; the corrected $L_V = 2(\alpha + \chi) = 4$ gives $\Delta t \leq 0.25$ and correctly forbids it. This is exactly why L_V must sum simultaneously-active weights (Assumption 2.5).

Effect on the theorems. A, C, D confirm the **necessity** of (respectively) the step-size condition, the driven caveat, and the loss-aware force — they do not invalidate the corrected D0 results. E refutes only the former Lipschitz constant and motivates the corrected L_V . B invalidates only the *extension* of the locality theorem to the sequential DE family, which is accordingly restricted to D0.

11. Exact scope and exclusions

The D0 theorems (4.4, 5.1, 5.2, 5.5, 5.6, 6.1, 8.1, 9.1) and Lemma 5.6a hold only for the synchronous, unconstrained, loss-aware, explicit-Euler law of Def 3.2. Each mechanism below is **excluded**, with the framework its rigorous treatment will likely need:

Excluded mechanism	Likely framework
clipping / projection at 0 and K	projected dynamical systems / variational inequalities
spill at K	one-sided projection / complementarity
unmet-demand saturation ($\min(d, \cdot)$)	nonsmooth / Filippov / piecewise-smooth analysis
hard-reserve constraints	constrained optimisation (KKT) / barrier methods
fixed activation cost c_0	hybrid / impulsive systems (discontinuous jumps)
golden-section / coordinate line search (DE24-family line-sized rules)	coordinate-descent theory (approximate 1-D minimisation); not proximal unless a proximal objective and proof are supplied
sequential live-state transfers	Gauss–Seidel operator splitting (vs Jacobi)
horizon optimisation	optimal control / dynamic programming
global / instantaneous field solves	elliptic PDE / implicit (nonlocal) solves
ledger incentives, EBU issuance	mechanism design — no dynamical-descent claim applies

Also excluded from D0 but present in the DE family and unaddressed here: **operator splitting** (drive N then transport A) and μ **frozen at $N(x^n)$** rather than x^n . These alone make D0 $\neq$ (any DE member) even before constraints; a Lie/Strang splitting-error analysis is the natural next tool.

12. Numerical validation (NOT proof)

The validation plan of the draft is implemented by `test_v28.py` (standard library only; deterministic seed 20260726; a self-contained D0 reference implementation, deliberately independent of every engine module). It contains **132 checks in 11 groups**, including an independently validated symmetric Jacobi eigensolver (analytic fixtures, eigenpair residuals, PSD and convergence guards) for the spectral bounds, and **four negative controls** that must fail if a corrected constant or scope restriction is silently reverted:

Group	Validates
1. marginals + curvature	analytic μ vs central differences on every smooth interval; branchwise slope; corrected exact and safe L_V (18 checks)
2. Counterexample E	negative control: the former max-form constant permits a V-increasing step; corrected L_V forbids it (8)
3. exact identities	Lemma 4.1 and Lemma 4.2 on active/inactive, lossy/lossless, $\theta = 0 / \theta > 0$ edges (11)
4. remainder inequality	$ r_n \leq R_n$ and the full (★) inequality, driven and undriven, cross terms, overlapping curvature (22)
5. one-edge bound + CE-A	144 deterministic sweep combos below the bound; exact tightness of the CE-A threshold (9)
6. spectral bound	Jacobi eigensolver validation; Gershgorin bound; descent at 0.95x the graph bound on seeded random graphs (19)
7. active-set + state-specific	Theorems 5.5/5.6; Lemma 5.6a ($J \neq 0 \Rightarrow SJ \neq 0$); $J = 0$ leaves the state exactly unchanged (22)
8. Counterexample D	negative control: executing the loss-blind transfer increases V; the D0 flux correctly stays zero (6)
9. stock/loss ledger	Theorem 8.1 to machine precision, lossless/lossy/mixed with drive (6)

10. locality	one-hop dependency under D0; negative control: a sequential live-state tick leaks to distance 2 (4)
11. $O(\Delta t^2)$ scaling	halving sequence: remainder bound holds; $ r_n /\Delta t^2$ bounded; $\sim 4\times$ shrinkage per halving (7)

Maximum residuals observed in the release run: $\max(|r_n| - R_n) = 2.4 \times 10^{-15}$ and max descent margin 7.1×10^{-15} (both pure floating-point roundoff at the exact-equality fixtures), max exact-identity residual 7.1×10^{-15} , max eigenpair residual 1.8×10^{-13} . The captured run is archived in `results/v2.8/v28_validation.txt`. **None of this is a proof:** the checks validate implementation consistency at the tested points only.

13. Unresolved conjectures and proof gaps

1. **Conjecture 5.7 (tightness)** — whether the active-set (5.5) or state-specific (5.6) sufficient bound is the tight admissible Δt is open; only the pure-quadratic one-edge case (Counterexample A) is settled.
2. **Driven global behaviour** — Corollary 6.2 is a one-step condition; multi-step boundedness/convergence under persistent drive (a discrete analogue of the coercive sublevel-set / LaSalle argument of V2.7 Cor 7.2) is **open**.
3. **Splitting error (D0 $\rightarrow$ DE family, step 1)** — bound the discrepancy from the engine's $A(N(\cdot))$ splitting with μ at $N(x^n)$ via Lie/Strang splitting analysis. **Open**.
4. **Loss-blind engine force** — DE-core and the DE24-family compute the loss-blind proposal force; characterise the set of states on which this is (non-)descending under $\eta < 1$ (Counterexample D is one witness). DE26-forced computes no force and is outside this item. **Open**.
5. **Constrained descent** — a projected-dynamics analogue of Theorem 4.4 admitting clipping/spill/reserve while retaining a dissipation inequality. **Open**.
6. **θ and the flat viable band** — the bounds drop the $\theta_e J_e$ term and ignore the flat ($v'' = 0$) interior; a curvature-aware, band-aware bound would be less conservative. **Open**.

14. Plain-language interpretation

Think of V as a 'stress score' for the whole grid and μ_i as the local pressure at cell i . Moving resource down a pressure gradient lowers stress — but along a lossy pipe only a fraction η arrives, so the *right* pressure difference to act on is $\mu_i - \eta\mu_j$, **not** $\mu_i - \mu_j$. Use the naive difference and you can 'help' a starving neighbour while wasting so much in transit that the grid is worse off (Counterexample D).

In continuous time (V2.7) stress falls at a rate equal to how hard the drive pushes minus how much the pipes dissipate. In discrete ticks you also pay a **step-size penalty**: take too big a step and you overshoot and *increase* stress (Counterexample A) — exactly like too large a learning rate in gradient descent. For one pipe the safe step is $\Delta t \leq 2/(L_V M(1+\eta^2))$; for a network it is governed by a spectral norm of the pipe layout. Physical *stock* is a separate ledger: transport never destroys resource mysteriously — whatever doesn't arrive is the named efficiency loss $(1 - \eta)$.

Two honest caveats. First, if the outside world keeps pumping resource in (drive), the score need not fall every tick; we can only say when dissipation beats drive plus the step penalty. Second — and most important — this all concerns the **idealised synchronous law (D0)**. The **real engine family (DE)** applies drive and transport in two ordered passes, moves resource one transfer at a time against a changing state (which can carry information two cells in a single tick), uses the naive loss-blind force wherever it computes a force at all (DE-core and the DE24-family; the forced harness DE26-forced just executes given actions), and clamps values to a physical range. **Proving D0 is a first brick, not the building: the engine's own guarantees still have to be earned separately.**

This note has not been peer reviewed. All theorems are proof attempts awaiting independent expert review; the numerical suite validates them at tested points only and proves nothing. No production engine, physics, or EBU-accounting behaviour is introduced or changed by V2.8.